$$e^0(41) = \frac{1,83,226}{10,543} = 17.37$$

Now, the completed Life Table becomes as

x	$l(x)$	$d(x)$	$p(x)$	$q(x)$	$L(x)$	$T(x)$	$e^0(x)$
40	10, 645	102	0.9905	0.0095	10,594	1, 93, 820	18.20
41	10, 543	169	0.9840	0.0160	10,459	1,83,226	17.37

Example 8.7

The following is a part of the Life Table constructed for a population, where the contents are incomplete. Evaluate the missing values using the given data and complete the Life Table.

x	$l(x)$	$d(x)$	$p(x)$	$q(x)$	$L(x)$	$T(x)$	$e^0(x)$
83	3560	-	-	0.16	-	-	
84	-	508	-	0.17	-	11975	

Solution:

Values of the missing entries can be estimated from the given data applying the respective formulae as follows:

The number of persons who die before reaching age of 83 years is calculated as

$d(83) = l(83) \times q(83)$

$= 3560 \times 0.16$

$= 569.6 = 570$

The value of the survivorship function $l(x)$ at $x = 84$ years is estimated as

$l(84) = l(83) - d(83)$

$= 3560 - 570$

$= 2990$

Values of $p(x)$ are estimated from the corresponding values of $q(x)$ as

$p(83) = 1 - q(83)$

$= 1 - 0.16 = 0.84$

$p(84) = 1 - q(84)$

$= 1 - 0.17 = 0.83$

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